

Performance improvements of commonly used optimizers in Machine Learning using **CUDA** or **OpenCL**

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Table of Contents

I. Motivation

II. Related Work & Expectations

III. Conceptual differences of OpenCL and CUDA

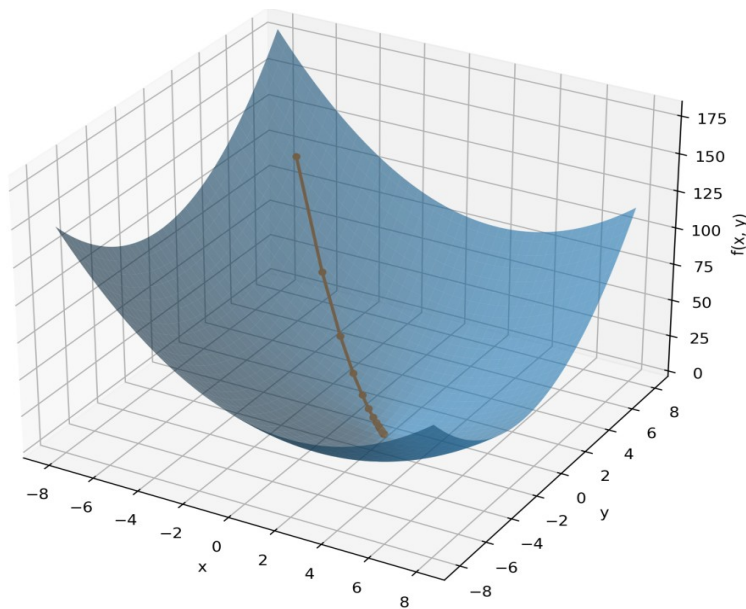
IV. The Implementation

V. Benchmarks

Motivation

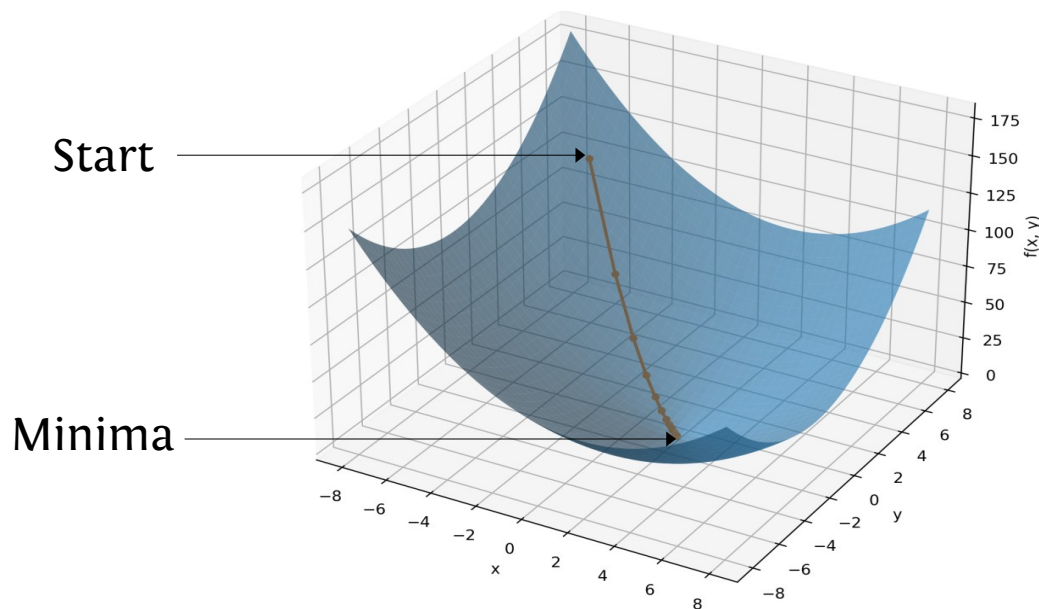
The role of optimization in machine learning

Optimizers are the engines of machine learning models.



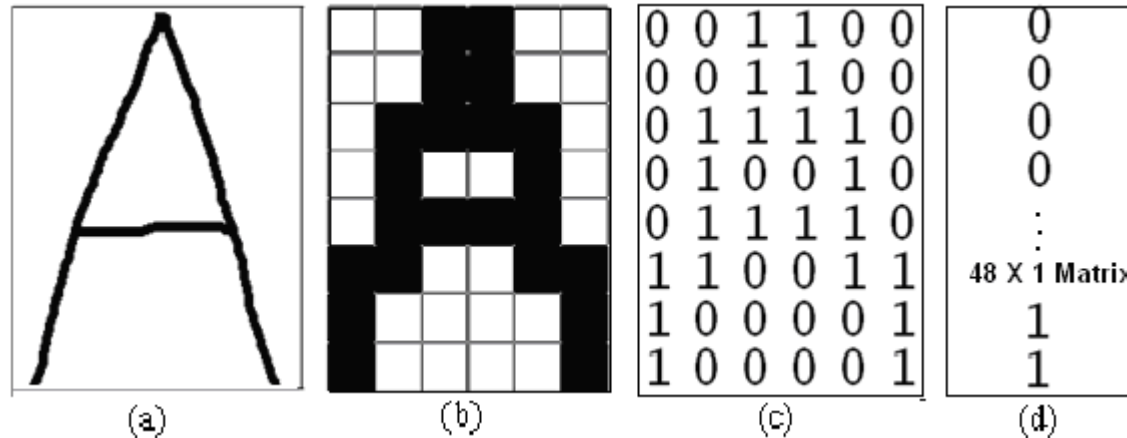
The role of optimization in machine learning

Train learnable weights towards a minimum of a loss function.



The role of optimization in machine learning

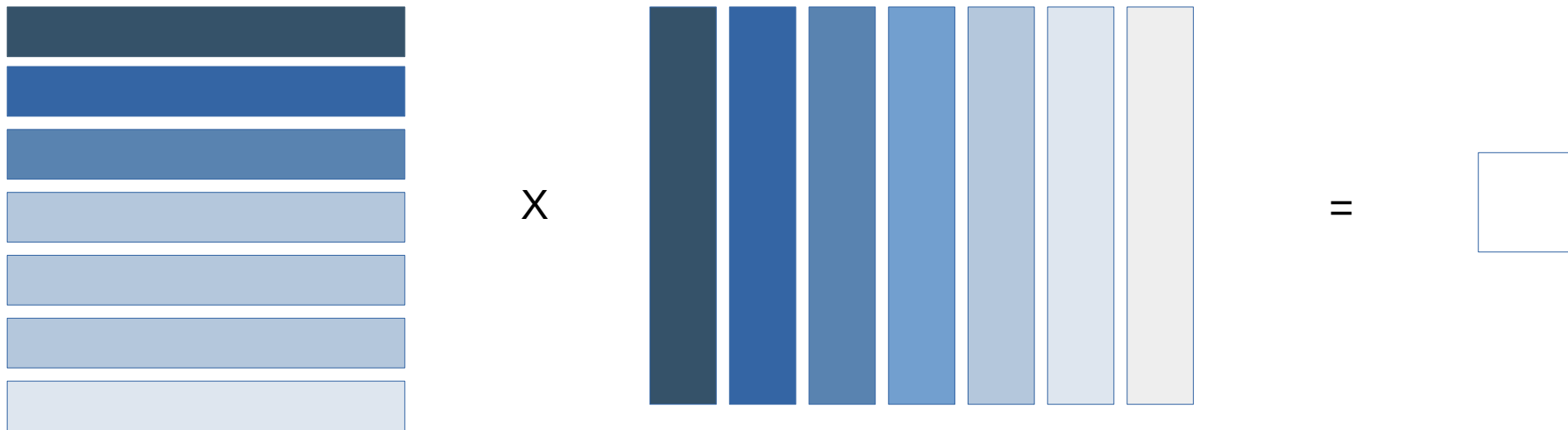
Most inputs i.e. images or text are represented as matrices...



https://www.researchgate.net/figure/a-Grayscale-image-of-character-A-b-Binary-representation-of-character-A-c_fig6_50211012

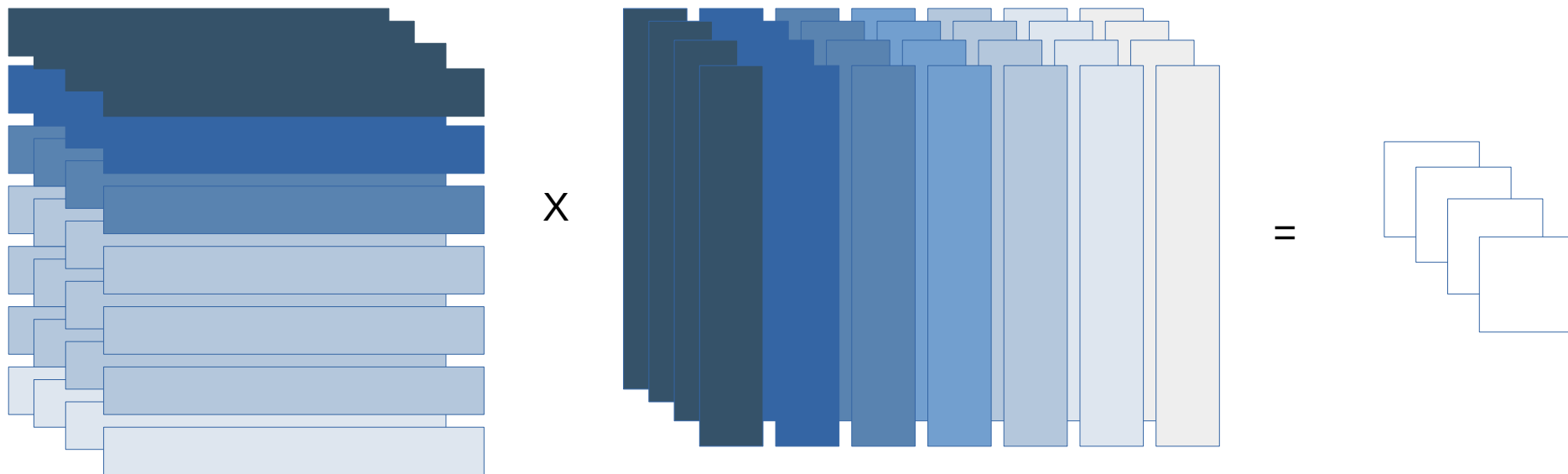
The role of hardware in machine learning

Thus we often deal with **matrix** or tensor operations during optimizations...



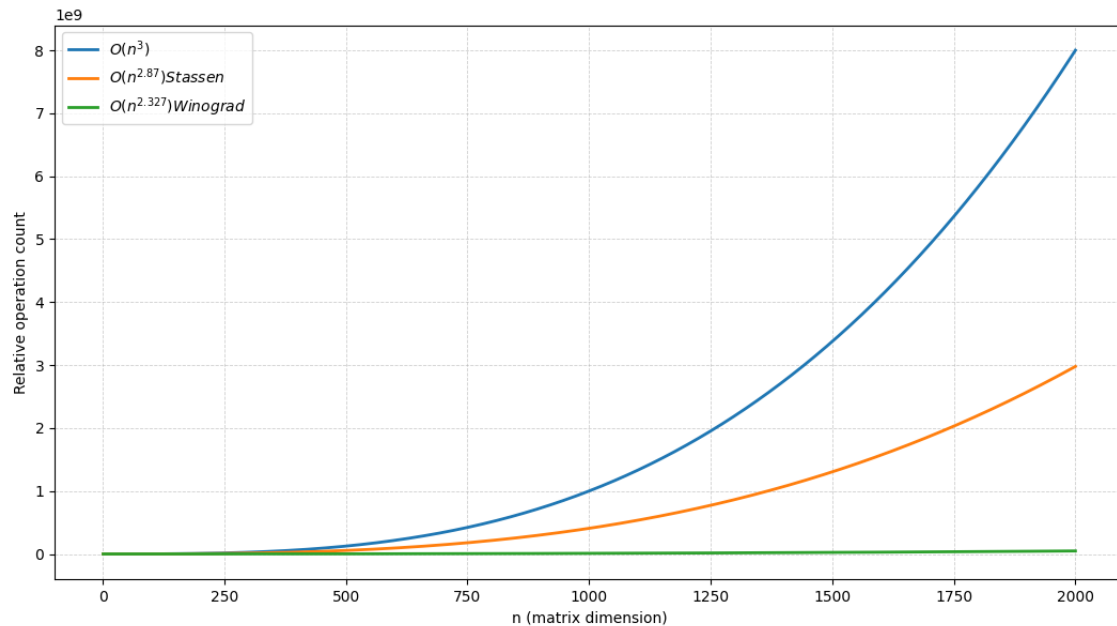
The role of hardware in machine learning

Thus we often deal with matrix or **tensor** operations during optimizations...



The role of hardware in machine learning

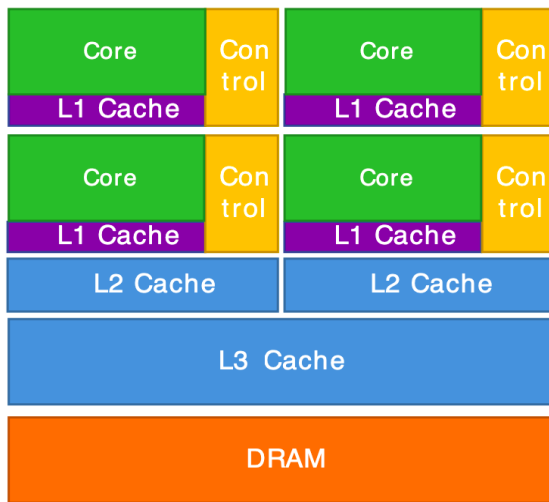
Matrix and Tensor operations are computationally expensive...



The role of hardware in machine learning

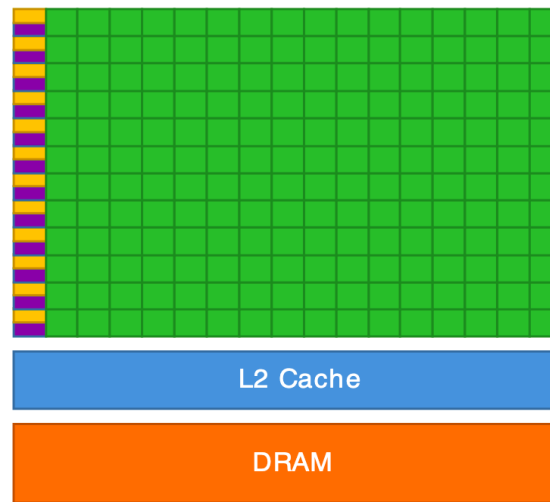
...exploiting hardware for parallelization

4 to 8 cores



CPU

100 to ~1000
cores



GPU

The role of hardware in machine learning

...how to exploit GPUs?

CUDA

OpenCL

Expectations

Results of „From CUDA to OpenCL“ [2]



Results of „A Comprehensive Performance Comparision of CUDA and OpenCL“ [2]

So what to expect from our benchmark?

Conceptual Comparision of CUDA and OpenCL

Architectural differences

Language & Usage differences

Benchmark program implementation

Architecture design

Workload Implementation

Optimizer Implementation in OpenCL

Optimizer Implementation in CUDA

Measurement and Benchmark Implementation

A Review of measured Benchmarks

Measured aspects in consideration

Visuals of obtained measurements

Conclusion so far

Outlook